

HEALTHCARE ANALYTICS

Concepts, Notes & Portfolio Guide

Project 1: CMS Hospital Quality Analysis

5,432 US Hospitals | MySQL · Excel · Power BI

Healthcare Informatics Portfolio

SECTION 1: SQL CONCEPTS

1. WHERE vs HAVING

Both are filters, but they run at different stages of a query.

WHERE: Filters individual rows BEFORE grouping. Use for row-level conditions like removing NULLs.

HAVING: Filters groups AFTER GROUP BY runs. Use for group-level conditions like minimum counts.

```
-- Only include states with at least 5 rated hospitals
GROUP BY state
HAVING COUNT(*) >= 5
```

Real-world connection: CMS uses this same logic — that is why you see the footnote 'Number of Cases Too Small' in raw CMS data. If a hospital does not have enough patients, CMS does not report a score because it would not be statistically meaningful.

2. Minimum Sample Size Threshold

In healthcare analytics, always filter by a minimum number of data points before calculating rates or averages. This is called a minimum sample size threshold.

```
HAVING COUNT(*) >= 5 -- at least 5 hospitals per state
```

A state with only 2 rated hospitals gives an unreliable average. A state where 1 out of 2 hospitals is penalized looks like a 50% penalty rate — but it is meaningless with such a small sample.

3. COUNT(*) vs COUNT(DISTINCT column)

One hospital appears many times in CMS data — once per measure. Knowing which COUNT to use is critical.

COUNT(*): Counts every row, including duplicates. Gives row count, not hospital count.

COUNT(DISTINCT facility_id): Counts each unique hospital only once, regardless of how many measures it appears in.

```
-- Wrong: counts rows (inflated)
SELECT COUNT(*) FROM complications_deaths;

-- Correct: counts unique hospitals
SELECT COUNT(DISTINCT facility_id) FROM complications_deaths;
```

4. Subquery

A subquery is a query inside another query. It runs first and returns a single value that the outer query uses.

```
ROUND(
  COUNT(DISTINCT facility_id) * 100.0 /
  (SELECT COUNT(DISTINCT facility_id) FROM complications_deaths),
  2) AS pct
```

The inner SELECT runs first, returns the total hospital count, then the outer query divides each group by that total to get a percentage.

5. Table Aliases in JOINS

When joining tables, give each a short nickname (alias) to avoid writing full table names repeatedly.

```
FROM hospital_info h
LEFT JOIN hac_reduction hac ON h.facility_id = hac.`Facility ID`
-- h = alias for hospital_info
-- hac = alias for hac_reduction
```

Aliases are required when two tables share column names (like facility_id). Writing h.facility_id vs hac.facility_id tells MySQL exactly which table you mean.

6. LEFT JOIN vs INNER JOIN

INNER JOIN: Only keeps rows that have a match in BOTH tables. Hospitals not in hac_reduction would be dropped entirely.

LEFT JOIN: Keeps ALL rows from the left table and brings in matching data from the right table where it exists. Non-matching rows show NULL for right-table columns.

```
FROM hospital_info h
LEFT JOIN hac_reduction hac ON h.facility_id = hac.`Facility ID`
-- Keeps all 5,432 hospitals even if not in HAC program
```

Use LEFT JOIN when you do not want to lose data just because one table is incomplete.

7. COALESCE

COALESCE(value, 0) means: if this value is NULL, use 0 instead.

```
-- Without COALESCE: NULL + 3 = NULL (breaks calculation)
-- With COALESCE:
COALESCE(count_of_mort_measures_worse, 0) +
COALESCE(count_of_safety_measures_worse, 0)
```

Used whenever doing math on columns that might have NULLs. Essential in CMS data where not all hospitals have all measures.

8. Backticks for Column Names with Spaces

When a column name has spaces, capital letters, or special characters, wrap it in backticks so MySQL reads it correctly.

```
hac.`Facility ID` -- correct
hac.Facility ID -- ERROR: MySQL reads ID as a separate word
```

This happens when data comes from different sources — Python import cleans names to lowercase_with_underscores, but MySQL Workbench wizard preserves original names with spaces. Always run DESCRIBE table_name before writing JOINS across tables from different sources.

9. LIKE with Wildcards

Used to filter text columns by pattern matching.

LIKE 'MORT%': Starts with MORT — matches MORT_30_AMI, MORT_30_HF, MORT_30_PN etc.

LIKE '%infection%': Contains the word infection anywhere in the string.

LIKE '_A%': Second character is A.

```
WHERE measure_id LIKE 'MORT%' -- isolates only mortality measures
```

10. Conditional Aggregation — SUM with CASE WHEN

Used to count a subset of rows within a group without a separate WHERE clause. One of the most commonly used patterns in real healthcare analytics.

```
SUM(CASE WHEN `Payment Reduction` = 'Yes' THEN 1 ELSE 0 END) AS penalized

-- Goes through every row:
-- Payment Reduction = Yes → add 1
-- Payment Reduction = No → add 0
-- SUM of all 1s = count of penalized hospitals
```

Why not just use WHERE? Because WHERE would remove the 'No' rows, making COUNT() only count penalized hospitals and losing your total. Conditional aggregation keeps all rows for the total while selectively counting the subset you care about.*

11. CAST — Converting Data Types

When data is loaded as text but you need to do math on it, use CAST to convert it to a number.

DECIMAL(10,2): For scores, rates, percentages — number with up to 10 digits and 2 decimal places.

UNSIGNED: For whole number counts.

DATE: For date strings.

```
-- avg_score is stored as text – CAST before using AVG()
ROUND(AVG(CAST(hcahps_linear_mean_value AS DECIMAL(10,2))), 2)
```

The Python import script loaded all columns as text (dtype=str) to prevent data type errors during loading. This means numeric columns must be CAST before any math.

12. Multiple LEFT JOINS

You can chain as many JOINS as needed. Each adds columns from another table. The key rule: always JOIN on the column linking the tables.

```
FROM hospital_info h
LEFT JOIN hac_reduction hac ON h.facility_id = hac.`Facility ID`
LEFT JOIN complications_deaths cd ON h.facility_id = cd.facility_id
-- facility_id is the PRIMARY KEY linking all CMS tables
```

13. ORDER BY Multiple Columns

Sorts by the first column first, then within ties sorts by the second column.

```
ORDER BY h.hospital_overall_rating DESC, -- 5-star hospitals first
hac.`Total HAC Score` ASC -- within same star rating, best HAC score first
```

Produces a meaningful ranked list — best hospitals at top, worst at bottom.

14. Multiple Aggregations in One Query

Calculate several metrics simultaneously in one SELECT — one pass through the data gives multiple insights at once.

```
SELECT hospital_ownership,  
AVG(hospital_overall_rating) AS avg_rating,  
COUNT(*) AS total_hospitals,  
SUM(CASE WHEN hospital_overall_rating=5 THEN 1 ELSE 0 END) AS five_star_count,  
ROUND(SUM(CASE WHEN hospital_overall_rating=5 THEN 1 ELSE 0 END)*100.0/COUNT(*),1) AS  
five_star_pct  
FROM hospital_info  
GROUP BY hospital_ownership;
```

SECTION 2: HEALTHCARE DATA CONCEPTS

CMS Hospital Compare — File Structure

CMS (Centers for Medicare & Medicaid Services) publishes quality data on every hospital in America across 4 separate files:

File	Rows	Key Columns	What it answers
Hospital General Info	5,432	Star rating, ownership, type, state	Which hospitals exist and their overall quality
Complications & Deaths	95,840	Measure ID, compared_to_national, score	How hospitals perform on mortality, safety, readmissions
HCAHPS Survey	325,856	HCAHPS question, linear mean, star rating	What patients think of their hospital experience
HAC Reduction Program	3,055	SIR values, Total HAC Score, Payment Reduction	Which hospitals are causing preventable infections and being penalized

The JOIN Key

Every CMS table links via **Facility ID** — a 6-digit unique hospital identifier. This is called the primary key/foreign key relationship. Every JOIN you write across CMS tables uses facility_id as the bridge.

compared_to_national — The Benchmarking Column

In the Complications & Deaths file, every hospital-measure combination is benchmarked against the national average:

Value	Meaning
Better Than the National Rate	Hospital rate is statistically significantly better than expected
No Different Than the National Rate	Hospital performs as expected — within normal variation
Worse Than the National Rate	Hospital rate is statistically significantly worse — a red flag
Number of Cases Too Small	Not enough patients to calculate a reliable rate (minimum sample size threshold)
NULL / Not Available	No data reported for this hospital-measure combination

CMS Measure Types

MORT — Mortality Measures

Track death rates for specific conditions within 30 days of hospital admission. The most serious quality indicators because they directly measure whether patients survive.

Examples: MORT_30_AMI (heart attack), MORT_30_HF (heart failure), MORT_30_PN (pneumonia), MORT_30_CABG (bypass surgery), MORT_30_COPD, MORT_30_STK (stroke)

HAI — Hospital-Acquired Infections

Infections patients get because they are in the hospital — not why they came in. Measured using SIR (Standardized Infection Ratio). SIR < 1.0 = better than expected. SIR > 1.0 = worse than expected.

Examples: CLABSI (central line infection), CAUTI (catheter UTI), SSI (surgical site infection), CDI (C. diff), MRSA (drug-resistant staph)

READM — Readmission Measures

Track 30-day unplanned readmission rates. High readmissions indicate patients are being discharged too early or without adequate discharge planning.

Examples: READM_30_AMI, READM_30_HF, READM_30_PN, READM_30_HIP_KNEE

PSI — Patient Safety Indicators

Surgery-related complications: blood clots, pressure ulcers, accidental cuts during surgery, sepsis after surgery. PSI 90 is a composite of all PSI measures.

Examples: Used in HAC Reduction Program scoring

Safety-Net Hospitals

Safety-net hospitals are public or non-profit hospitals that serve the highest proportion of uninsured, low-income, and undocumented patients regardless of ability to pay. They are chronically underfunded and treat the sickest patients with the fewest resources.

Key insight: Safety-net hospitals almost always rank lower on CMS quality measures — but this does NOT mean bad doctors or bad care. It reflects structural disadvantage: less funding, sicker patients, fewer specialists. Raw quality scores without socioeconomic context can lead to misleading conclusions and unfair penalization of hospitals doing the hardest work.

Rate vs Count — A Critical Distinction

When comparing states or groups, always clarify whether you are reporting a rate (percentage) or a count (raw number).

Metric	Best for	Example from this project
COUNT (raw number)	Understanding absolute scale. Resource allocation	California has 76 penalized hospitals — most in the US
RATE (percentage)	Comparing states fairly regardless of size. Policy decisions	California has 67% penalty rate — highest proportion in the US

Rule: Always clarify which one you are reporting and why. Reporting only one without context can be misleading.

SECTION 3: KEY FINDINGS FROM THE DATA

National Mortality Landscape

76.54% (3,668 hospitals): Perform no differently than the national average on mortality

10.16% (487 hospitals): Perform BETTER — saving more lives than expected

7.85% (376 hospitals): Perform WORSE — losing more patients than expected

Geographic Quality Patterns

Best states: SD, CO, UT: Strong rural hospital networks, high healthcare spending, healthier populations

Worst states: PR, MS: Puerto Rico — underfunded, Hurricane Maria damage. Mississippi — poverty, chronic disease, rural isolation

Consistent pattern: Mississippi appears in BOTH worst star ratings AND worst individual hospitals — systemic inequality, not coincidence

Hospital Ownership vs Quality

Highest: Veterans Health Administration: Federal standardization, no profit motive, pioneered EHR (VistA), dedicated patient population

Lowest: Proprietary (for-profit): Shareholder obligations, less nursing staff investment, cost-cutting, avoid unprofitable service lines

Non-profit outperforms for-profit: Confirmed across 3,000+ hospitals — real evidence in a decades-old policy debate

Patient Experience (HCAHPS)

Highest scores: Nurse Communication (91), Doctor Communication (91): Direct clinical interactions rated highly — face-to-face care works

Lowest scores: Quietness (81), Communication about Medicines (77): Structural problems — hospital noise, medication explanation gaps

Critical finding: Discharge Information scores low: Patients leave without understanding follow-up care — directly causes preventable readmissions

HAC Penalty Program

23.5% of hospitals penalized: Close to the designed 25% bottom quartile — program working as intended

Highest penalty RATES: VT, HI, IA: Small states — few hospitals so rate is disproportionately high

Highest penalty COUNTS: CA, NY, PA: Large states — more hospitals means more absolute penalties

Best Individual Hospitals

Lebanon VA Medical Center, Philadelphia VA Medical Center: VA hospitals dominate the top — standardization wins

Waverly Health Center (Iowa): Small critical access hospital — tight-knit staff, low volume, high accountability

Key insight: Fame and size do not equal quality — data reveals best outcomes in smaller, less well-known institutions

SECTION 4: POWER BI CONCEPTS

Data Model and Relationships

Power BI loads each CSV as a separate table. Without relationships, a slicer from one table cannot filter visuals from another table. Relationships must be manually drawn in Model View by dragging matching columns between tables.

The common key across all CMS state-level tables is the state column. `facility_id` links hospital-level tables. Always identify your join key before building the data model.

Slicers and Cross-Filtering

Slicers filter visuals on the same page. A slicer only filters visuals that use the same table OR tables connected via relationships in the data model.

Tables without a common column (like Q3 mortality — national totals only, no state) will remain static regardless of slicer selection. This is expected and correct behaviour.

Page Background vs Wallpaper

Wallpaper: shows on screen only — does NOT export to PDF. Page Background: shows on screen AND exports to PDF. Always use Page Background when you want the colour to appear in exports.

CAST / Data Types in Power Query

CSVs load all columns as text by default. Numeric columns must be converted in Power Query (Transform Data) before they can be used in visual field wells like Color Saturation.

Always check column icons in the Data panel — ABC = text, 123 = number. Change data type in Power Query before building visuals.

Conditional Formatting for Maps

Newer Power BI versions removed Color Saturation from Filled Map field wells. Instead apply color via: Format tab → Data Colors → fx button → Format by Field Value → set min/max colors.

Set minimum = 1 (Red) and maximum = 5 (Green) to span the full scale rather than auto-scaling to your data range, which compresses colour differences.

Slicer Types

Dropdown: best for long lists (50 states). Tile: best for short fixed categories (star ratings 1-5). List: best for medium-length categorical data.

SECTION 5: PORTFOLIO & CAREER GUIDE

Resume / LinkedIn Bullet Points

- Analyzed CMS Hospital Compare data across 5,432 US hospitals using MySQL, Excel, and Power BI to identify quality gaps in American healthcare
- Built end-to-end analytics pipeline: raw government data (500,000+ rows across 4 files) → SQL cleaning and analysis → Power BI interactive dashboard
- Identified that 376 hospitals (7.85% nationally) have mortality rates worse than the national average through SQL analysis of CMS Complications & Deaths data
- Revealed geographic disparities in hospital quality across all 50 states — Puerto Rico and Mississippi consistently rank lowest across multiple independent quality measures
- Demonstrated that for-profit hospitals score 15-20% lower on CMS star ratings than non-profit hospitals through ownership-type analysis of 3,000+ facilities
- Applied HCAHPS patient experience data (325,856 survey records) to identify discharge communication as the lowest-scoring care domain nationally

STAR Interview Answer

SITUATION: CMS publishes quality data on every hospital in America, but it is spread across 4 separate files with over 500,000 rows total — making it difficult to extract actionable insights without a structured analytical approach.

TASK: Design and execute an end-to-end analytics pipeline to surface meaningful insights about hospital quality disparities across the United States.

ACTION: Loaded all 4 CMS files into MySQL using a Python ETL script that handled encoding issues and NULL conversion. Wrote 8 analytical SQL queries using JOINS across tables, subqueries, conditional aggregation, and CAST for data type conversion. Built a Power BI dashboard with 5 visuals and cross-filtering slicers connected via a data model with table relationships.

RESULT: Identified that 376 hospitals (7.85% nationally) have worse-than-national mortality rates, that VA hospitals consistently outperform for-profit hospitals across all quality dimensions, and that discharge communication is the single lowest-scoring patient experience domain — with direct implications for preventable readmission rates.

Personal Statement Angles

As a BDS graduate, you bring something most healthcare informatics students do not have: you understand what these metrics mean clinically. When you see a low discharge communication score in HCAHPS data, you know from clinical experience that patients who do not understand their post-treatment instructions are at real risk. When you see HAI rates, you understand infection control from a practitioner's perspective. This clinical grounding makes your data insights more meaningful and your career trajectory more compelling.

Suggested personal statement framing:

"Analysis of 325,856 HCAHPS survey responses revealed that while patients rate nurse and doctor communication highly, discharge information and medication communication consistently score lowest nationally. As a dental clinician who has experienced the communication gap between provider and patient

firsthand, I want to build the informatics systems that close this gap — because the data proves this gap directly causes preventable readmissions and patient harm."

Project Summary for Portfolio

Item	Detail
Project Name	CMS Hospital Quality Analysis
Data Source	Centers for Medicare & Medicaid Services (CMS) Hospital Compare
Dataset Size	500,000+ rows across 4 files, 5,432 hospitals
Tools Used	MySQL, Excel, Power BI, Python (ETL)
SQL Techniques	JOINS, Subqueries, CASE WHEN, COALESCE, CAST, GROUP BY, HAVING, LIKE
Key Findings	376 hospitals with worse mortality, VA hospitals outperform for-profit, discharge communication lower
Dashboard	5 visuals: State map, HAC penalty chart, Mortality donut, Ownership bar, HCAHPS survey chart